WORKED SOLUTION

STEP III 1990, Question 4

Pure • Trigonometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q4

The key idea

Both series are sums of cosines whose angles go up in equal steps, so there are two natural tools. For the plain sum, multiply by $2\sin\beta$ and use a product-to-sum formula: every term splits into a difference of two sines and the sum telescopes. For the binomial sum, and for the final identity, think of $\cos\phi$ as the real part of $e^{i\phi}$. Then the binomial theorem or the geometric-series formula does the work, and the answer comes out by taking real or imaginary parts at the end.

Solution

The first series

Call the sum S. By the product-to-sum formula $2\sin A\cos B = \sin(B+A) - \sin(B-A)$,

$$2\sin\beta\cos(\alpha+2r\beta) = \sin(\alpha+(2r+1)\beta) - \sin(\alpha+(2r-1)\beta).$$

Adding these for $r = 0, 1, \ldots, n$, the sum telescopes: the first sine for r cancels the second sine for $r+1$. So only two terms are left:

$$\begin{aligned} 2\sin\beta\, S &= \sin(\alpha+(2n+1)\beta) - \sin(\alpha-\beta) \\ &= 2\cos(\alpha+n\beta)\sin((n+1)\beta), \end{aligned}$$

using $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$ with $\frac{P+Q}{2} = \alpha+n\beta$ and $\frac{P-Q}{2} = (n+1)\beta$. Since $\sin\beta \neq 0$ we may divide, and

$$\boxed{\sum_{r=0}^{n}\cos(\alpha+2r\beta) = \frac{\sin(n+1)\beta\,\cos(\alpha+n\beta)}{\sin\beta}}$$

Notice that $\alpha+n\beta$ is the average of the first and last angles, which is a good sanity check on the form.

The binomial series

Call this sum T. Since $\cos\phi = \operatorname{Re} e^{i\phi}$ and the binomial coefficients are real,

$$T = \operatorname{Re} \sum_{r=0}^{n} \binom{n}{r} e^{i\alpha}(e^{2i\beta})^r = \operatorname{Re}\left[e^{i\alpha}(1+e^{2i\beta})^n\right]$$

by the binomial theorem. Now take out the "half-angle" factor:

$$1 + e^{2i\beta} = e^{i\beta}(e^{-i\beta} + e^{i\beta}) = 2\cos\beta\, e^{i\beta}.$$

Hence $e^{i\alpha}(1+e^{2i\beta})^n = 2^n \cos^n\beta\, e^{i(\alpha+n\beta)}$, and since $2^n\cos^n\beta$ is real,

$$\boxed{\sum_{r=0}^{n} \binom{n}{r} \cos(\alpha + 2r\beta) = 2^n \cos^n\beta\, \cos(\alpha + n\beta)}$$

(This one doesn't actually need $\sin\beta \neq 0$.)

The final identity

The secants mean we need $\cos\theta \neq 0$, which is implicit in the question. The rth term is $\cos r\theta\, \sec^r\theta = \operatorname{Re}(e^{i\theta}\sec\theta)^r$, because $\sec^r\theta$ is real. So put

$$w = e^{i\theta}\sec\theta = \frac{\cos\theta + i\sin\theta}{\cos\theta} = 1 + i\tan\theta.$$

The left-hand side is $\operatorname{Re}(1 + w + \cdots + w^n)$.

> **Ask yourself** Why is it safe to use the geometric-series formula here, and why does w help?

The formula needs $w \neq 1$, and $w - 1 = i\tan\theta \neq 0$ precisely because $\sin\theta \neq 0$. That's where the condition is used. The point of w is that $w - 1$ is purely imaginary, so dividing by it is easy, while w^{n+1} is easy to compute from the exponential form $w = \sec\theta\, e^{i\theta}$. Writing $t = \tan\theta$,

$$1 + w + \cdots + w^n = \frac{w^{n+1} - 1}{w - 1} = \frac{\sec^{n+1}\theta\, e^{i(n+1)\theta} - 1}{it}.$$

For any complex z and real $t \neq 0$ we have $\operatorname{Re}\dfrac{z}{it} = \operatorname{Re}\dfrac{-iz}{t} = \dfrac{\operatorname{Im} z}{t}$. The -1 in the numerator has zero imaginary part, so

$$\begin{aligned}\sum_{r=0}^{n} \cos r\theta\, \sec^r\theta &= \frac{\sec^{n+1}\theta\, \sin(n+1)\theta}{\tan\theta} = \sec^{n+1}\theta\, \sin(n+1)\theta \cdot \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin(n+1)\theta\, \sec^n\theta}{\sin\theta},\end{aligned}$$

which is the required result.

Link with the first part. Multiplying through by $\cos^n\theta$, the identity says

$$\sum_{r=0}^{n} \cos r\theta \cos^{n-r}\theta = \frac{\sin(n+1)\theta}{\sin\theta}.$$

You can also prove it in that form by induction on n: if C_n is the left-hand side then $C_n = \cos\theta\, C_{n-1} + \cos n\theta$, and $\sin n\theta\cos\theta + \cos n\theta\sin\theta = \sin(n+1)\theta$ finishes the step.

Checks

- $n = 0$: all three formulae reduce to their single first term ($\cos\alpha$, $\cos\alpha$ and 1).
- In the first sum, $\beta \to 0$ gives $\frac{\sin(n+1)\beta}{\sin\beta} \to n+1$, so $S \to (n+1)\cos\alpha$, which is right for $n+1$ equal terms. In the second, $\beta = 0$ gives $2^n\cos\alpha$, since $\sum \binom{n}{r} = 2^n$.
- $\theta = \pi/4$, $n = 2$: the left-hand side is $1 + \frac{1}{\sqrt{2}}\cdot\sqrt{2} + 0 = 2$, and the right-hand side is $\sin\frac{3\pi}{4}\cdot 2/\sin\frac{\pi}{4} = 2$.
- Numerically, all three formulae agree with direct summation to within 10^{-9} (relative) over 20 000 random choices of α, β, θ and $n \le 15$.

Common slips

- Forgetting that the first series has $n+1$ terms ($r = 0$ to n), which gives $\sin n\beta$ instead of $\sin(n+1)\beta$.
- Writing $1 + e^{2i\beta}$ as $2\cos\beta$ and losing the factor $e^{i\beta}$, which shifts the angle in the answer.
- In the last part, taking the real part of the numerator and the denominator separately. You must first make the denominator real.

Beyond STEP

Writing cos as the real part of an exponential turns trigonometric sums into geometric or binomial sums. This is the starting point of Fourier analysis: the first sum is essentially the Dirichlet kernel $\sin\big((n+\frac{1}{2})x\big)/\sin\frac{x}{2}$, which controls how Fourier series converge. Needham's *Visual Complex Analysis* gives a beautifully geometric view of why multiplying by $e^{i\theta}$ means rotation, and why sums like these close up into neat formulae.

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